Tier	Prediction	Measurement protocol	Target effect size	Falsification criterion	Feasibility horizon	Validation status
Tier 3 Computational	P1: Nicotinic agonist lowers effective threshold	Nicotine or α4β2 agonist vs placebo; psychophysics (d′) and EEG (P3b amplitude)	d′ increase ≥ 0.3; P3b amplitude increase ≥ 20%	No change in detection/P3b.	now	Theoretically specified
Tier 3 Computational	P2: Near-threshold bistability with critical slowing	Staircase near threshold; trial-to-trial RT/EEG; estimate lag-1 autocorrelation (AC1)	AC1 increase ≥ 0.2 within 10% of threshold	A continuous graded relationship.	now	Theoretically specified
Tier 3 Computational	P3: 200–300 ms frontoparietal TMS abolishes report, spares priming	TMS at 200–300 ms over frontoparietal cortex; measure report (d′) and priming	Report d′ drop ≥ 0.8; priming effect preserved (≤ 20% reduction)	Both disrupted equally or access persists.	now	Theoretically specified
Tier 3 Computational	P4: $S_t > \theta_t$ predicts access at d′ ≥ 0.5	Single-trial decoding of S_t and θ_t (EEG/iEEG); predict access on held-out trials	AUC ≥ 0.80; accuracy improvement ≥ 10% over chance	d′ < 0.5 (N ≥ 2,000).	now	Theoretically specified
Tier 2 Info-Theoretic	P5: Attentional cueing raises mutual information by ≥1 bit	Cue vs no-cue; estimate stimulus–brain mutual information (MI)	ΔMI ≥ 1 bit	No increase.	2–5 years	Pending
Tier 2 Info-Theoretic	P6: Conscious bandwidth asymptotes ~40 bits/s	Rapid RSVP across modalities; estimate information rate (bits/s) vs stimulation rate	Asymptote = 30–50 bits/s; fit saturating curve	Modality variation or sustained > 100 bits/s.	2–5 years	Pending
Tier 2 Info-Theoretic	P7: θ_t approximates the Neyman–Pearson optimum	Model-based ROC analysis; compare θ_t to NP-optimal criterion	Criterion within 2 SD of NP optimum	Deviation > 2 SD.	2–5 years	Pending
Tier 2 Info-Theoretic	P8: Interoceptive transfer entropy > exteroceptive by ≥ 0.05 bits/trial	iEEG/EEG; compute TE from interoceptive vs exteroceptive regions to frontoparietal cortex	ΔTE ≥ 0.05 bits/trial	TE_intero ≤ TE_extero.	2–5 years	Pending
Tier 1 Thermodynamic	P9: Ignition trials cost more frontoparietal energy ($CMRO_2$/glucose)	Simultaneous fMRI/PET ($CMRO_2$ or glucose); compare ignition vs matched sub-threshold trials	$\Delta CMRO_2$ or Δglucose ≥ 10% after controls	No differential after controls.	5–10 years	Pending
Tier 1 Thermodynamic	P10: Threshold-gated processing more energy-efficient than continuous	Adaptive (threshold-gated) vs continuous stimulation/task; energy per bit or per correct report	≥ 20% lower energy cost	Continuous matches or beats it.	5–10 years	Pending
Tier 1 Thermodynamic	P11: Metabolic depletion elevates θ_t	Induce metabolic depletion (hypercapnia/ hypoxia or TMS); measure θ_t via staircases	θ_t increase ≥ 0.3 SD	No effect or lowered threshold.	5–10 years	Pending
Tier 1 Thermodynamic	P12: Single-ignition cost exceeds the Landauer minimum by κ (≈100 ATP/bit)	Estimate energy cost per ignition (ATP equivalents); compare to Landauer limit ($k_BT \ln 2$)	κ ≥ 100 ATP/bit ($\sim 4\times10^{-20}$ J/bit)	κ below the canonical floor or BOLD/direct diverge > 10×.	5–10 years	Pending

Validation status
- Theoretically specified
- Pending